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ZAKLJUČNA NALOGA (FINAL PROJECT PAPER)

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UNIVERZA NA PRIMORSKEM
FAKULTETA ZA MATEMATIKO, NARAVOSLOVJE IN INFORMACIJSKE TEHNOLOGIJE

ZAKLJUČNA NALOGA
(FINAL PROJECT PAPER)

PODATKOVNO VODENA IDENTIFIKACIJA NABOROV NAPOVEDNIH FUNKCIJ ZA KLASIFIKACIJO SINDROMA POLICISTIČNIH JAJČNIKOV
(DATA-DRIVEN IDENTIFICATION OF PREDICTIVE FEATURE SETS FOR POLYCYSTIC OVARIAN SYNDROME CLASSIFICATION)

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Zaključna naloga
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Podatkovno Vodena Identifikacija Naborov Napovednih Funkcij za Klasifikacijo Sindroma Policističnih Jajčnikov

(Data-Driven Identification of Predictive Feature Sets for Polycystic Ovarian Syndrome Classification)

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Izvleček:
Sindrom policističnih jajčnikov (PCOS) je pogosta endokrina motnja, ki prizadene ženske v reproduktivnem obdobju. Povezan je s širokim spektrom znakov in simptomov, vključno z menstrualnimi nepravilnostmi, debelostjo, neplodnostjo in presnovnimi zapleti. Natančna diagnoza ostaja zahtevna, saj je PCOS heterogena motnja in se lahko kaže z različnimi kombinacijami simptomov, biokemičnih označevalcev ter ultrazvočnih ugotovitev. Ta zaključna naloga raziskuje uporabo strojnega učenja za klasifikacijo PCOS z uporabo podatkovno podprte identifikacije napovednih naborov značilk.

Zato ta raziskava preučuje, kateri nabori diagnostičnih informacij zagotavljajo najmočnejšo napovedno uspešnost pri klasifikaciji PCOS. Razpoložljivi napovedniki so bili razvrščeni glede na stopnje diagnostičnih informacij, vključno s hormonskimi, krvnimi, pacientu opaznimi, zdravniku opaznimi, ginekološkimi in celotnimi nabori značilk. Ti nabori značilk so bili ovrednoteni z uporabo dveh komplementarnih eksperimentalnih pristopov: primerjave postopnega dodajanja naborov značilk in ablacijske analize. Uporabljeni so bili trije modeli strojnega učenja: logistična regresija, naključni gozd in umetna nevronska mreža.

Rezultati kažejo, da je sestava nabora značilk pomembno vplivala na uspešnost klasifikacije. Ginekološki nabor značilk je dosegel najboljše rezultate med nabori v postopni primerjavi, medtem ko so hormonski in krvni nabori značilk pri samostojni uporabi dosegli najslabše rezultate. Ablacijska analiza je pokazala, da je odstranitev ginekoloških informacij povzročila največji upad uspešnosti, medtem ko odstranitev hormonskih ali krvnih spremenljivk ni zmanjšala uspešnosti modelov. Analiza pomembnosti značilk z uporabo naključnega gozda je dodatno pokazala, da so bili med najvplivnejšimi napovedniki število foliklov in simptomi, ki jih lahko opazi pacientka. Analiza lažno pozitivnih in lažno negativnih primerov je prav tako pokazala, da so nekateri modeli dosegli visoko občutljivost (recall), vendar na račun povečanega števila lažno pozitivnih napovedi.

Naloga ugotavlja, da različne vrste diagnostičnih informacij ne prispevajo enako h klasifikaciji PCOS. Ginekološke informacije, zlasti morfologija jajčnikov, so bile v tej podatkovni zbirki najbolj napovedne, medtem ko so tudi simptomi, ki jih lahko opazi pacientka, zagotavljali uporabne napovedne informacije. Te ugotovitve kažejo, da lahko strukturirano vrednotenje naborov značilk izboljša interpretabilnost modelov za klasifikacijo PCOS in pomaga določiti, katere diagnostične informacije so najvrednejše za podatkovno podprto odkrivanje PCOS. Prihodnje raziskave bi morale te ugotovitve potrditi na večjih in bolj raznolikih podatkovnih zbirkah ter raziskati različne tehnike in metode za klinično uporabo.

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Abstract:
Polycystic Ovarian Syndrome (PCOS) is a common endocrine disorder affecting women of reproductive age. It is associated with a wide range of signs and symptoms, including menstrual irregularities, obesity, infertility, and metabolic complications. Accurate diagnosis remains challenging because PCOS is heterogeneous and may present through different combinations of symptoms, biochemical markers, and ultrasound findings. This thesis explores the use of machine learning for PCOS classification through the data-driven identification of predictive feature sets.

Therefore, this study investigates which diagnostic-information feature sets provide the strongest predictive performance for PCOS classification. The available predictors were grouped according to stages of diagnostic information, including hormonal, blood-test, patient-observable, doctor-observable, gynecological, and all-feature sets. These feature sets were evaluated using two complementary experimental approaches: incremental feature-set comparison and ablation analysis. Three machine learning models were used: Logistic Regression, Random Forest, and Artificial Neural Network.

The results show that feature-set composition substantially affected classification performance. The gynecological feature set achieved the strongest performance among the incremental feature sets, while hormonal and blood-test feature sets performed the weakest when used alone. The ablation analysis showed that removing gynecological information caused the largest performance decline, whereas removing hormonal or blood-test variables did not reduce performance. Feature importance analysis using Random Forest further showed that follicle counts and patient-observable symptoms were among the most influential predictors. False positive and false negative analysis also showed that some models achieved high recall at the cost of increased false positives.

The thesis concludes that different types of diagnostic information do not contribute equally to PCOS classification. Gynecological information, particularly ovarian morphology, was the most predictive in this dataset, while patient-observable symptoms also provided useful predictive information. These findings suggest that structured feature-set evaluation can improve the interpretability of PCOS classification models and help identify which diagnostic information is most valuable for data-driven PCOS detection. Future work should validate these findings on larger and more diverse datasets and explore different techniques and methods for clinical use.

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First and foremost, I would like to express my sincere gratitude to my mentor for guiding me throughout this entire process. His patience, support, and willingness to answer every question I had were invaluable and greatly contributed to the completion of this thesis.

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List of Abbreviations

PCOS	Polycystic Ovary Syndrome
NIH	National Institutes of Health
ML	Machine Learning
AI	Artificial Intelligence
LR	Logistic Regression
RF	Random Forest
ANN	Artificial Neural Network
AMH	Anti-Müllerian Hormone
AUC	Area Under the Curve

1 Introduction

Polycystic Ovarian Syndrome (PCOS) is a prevalent endocrine disorder affecting women of reproductive age, marked by a wide range of signs and symptoms, including menstrual disturbances, obesity, hirsutism, and androgenic alopecia [5]. The main characteristics of PCOS include elevated androgen levels and the presence of multiple follicles that may be detected via ultrasound examination. However, despite the term "polycystic" in the name, the condition can be diagnosed even in the absence of ovarian cysts.

Epidemiological studies estimate that PCOS impacts approximately 10–13% of reproductive-aged women. Despite its high prevalence, it is estimated that globally up to 70% of women with PCOS are unaware that they have this condition [1]. Diagnosis typically occurs in patients' 20s or 30s, frequently during fertility tests rather than through earlier clinical screening. Delayed or inaccurate diagnosis may contribute to the progression of other health issues such as insulin resistance, type 2 diabetes, obesity, and endometrial cancer [10]. Therefore, PCOS is increasingly recognized not only as a reproductive disorder but also as a broader metabolic condition with long-term clinical consequences.

Despite its prevalence, the diagnosis of PCOS remains challenging due to the heterogeneous nature of the syndrome and the absence of a universally accepted diagnostic standard. Diagnosis is currently most commonly guided by the Rotterdam criteria, where PCOS is defined by the presence of two out of the following three criteria: oligo-anovulation, hyperandrogenism, and polycystic ovarian morphology [14].

The Rotterdam criteria are relevant to this thesis because they demonstrate that the PCOS diagnosis is based on multiple types of diagnostic information. Hyperandrogenism is often diagnosed using hormonal and biochemical tests, while polycystic ovarian morphology is evaluated using an ultrasound examination. In addition, many women exhibit visible signs such as menstrual irregularities, weight gain, or excessive hair growth. Therefore, PCOS diagnosis relies on a combination of patient-observable, clinical, biochemical, and ultrasound-based information rather than a single diagnostic indicator.

Earlier definitions, such as those from the NIH (National Institutes of Health),

[1] World Health Organization. *Polycystic Ovary Syndrome*, 2023. Available at: `https://www.who.int/news-room/fact-sheets/detail/polycystic-ovary-syndrome`.

were more restrictive because they required both hyperandrogenism and ovulatory dysfunction.

This distinction is important because the Rotterdam criteria include polycystic ovarian morphology detected by ultrasound, allowing for diagnosis even in the absence of hyperandrogenism. Because of this broader definition and the inclusion of ultrasound, the reported prevalence of PCOS has increased. However, ultrasound findings can sometimes lead to false positives, as polycystic ovarian morphology may appear in healthy women and during adolescence. Therefore, reliable PCOS diagnosis often depends on combining multiple types of clinical, biochemical, and imaging-based information rather than relying on a single diagnostic indicator [13].

While these information sources are frequently combined in clinical practice, their relative contribution to diagnosis and prediction remains less clear. As a result, there is limited understanding of which types of diagnostic information provide the greatest predictive value and whether some information sources contribute more strongly to PCOS classification than others.

Because the diagnosis of PCOS relies on multiple imperfect and heterogeneous indicators, and because the relative predictive value of these information sources remains uncertain, researchers have increasingly explored machine learning (ML) and artificial intelligence (AI) approaches to support earlier and more consistent detection. The growing adoption of data-driven methods in healthcare has further motivated the application of ML techniques to PCOS prediction.

Numerous studies have examined the performance of several ML models, such as Logistic Regression (LR), Random Forest (RF), and K-nearest neighbors, for PCOS prediction. However, despite extensive model comparison, limited attention has been given to the systematic evaluation of feature sets contributions [15].

Therefore, this thesis aims to examine the predictive value of different feature groups and their contribution to PCOS classification performance. The study is guided by the following research question:

Which diagnostic information feature sets, including hormonal, blood-test, patient-observable, doctor-observable, and gynecological feature sets, provide the strongest predictive performance for PCOS classification?

This question is examined both by training models on individual feature sets and by performing ablation experiments in which each set is removed from the full feature set.

2 Related Work

This chapter reviews existing literature on PCOS diagnosis and machine learning-based prediction. It begins by outlining the evolution of PCOS diagnostic criteria because the construction of PCOS datasets is directly impacted by differences in diagnostic definitions. Following a discussion of machine learning techniques for PCOS prediction, including preprocessing, feature selection, and model optimization. This research collectively serves as the foundation for this thesis, which compares the predictive value of significant feature sets.

2.1 Evolution of PCOS Diagnostic Criteria

Since the identification of PCOS, there has been no universally accepted diagnostic standard. The earliest guideline was initiated by the National Institutes of Health (NIH) in 1990, where the diagnosis was clinical and/or biochemical; hyperandrogenism was combined with ovulatory problems, and ultrasound was suggestive, not necessary.

However, this definition differed from European clinical practice at the time, where ultrasound findings were a definitive feature in the analysis. To address these inconsistencies, 27 specialists met in Rotterdam in 2003, resulting in the widely adopted Rotterdam criteria [3]. These criteria are explained by four phenotypes based on combinations of hyperandrogenism (H), ovulatory dysfunction (O), and polycystic ovarian morphology (P), as summarized below [18]:

Table 1: Rotterdam PCOS phenotypes

Phenotype	**Criteria**
A	H + O + P
B	H + O
C	H + P
D	O + P

H: Hyperandrogenism (elevated androgen levels)

O: Oligo/anovulation (irregular or absent ovulation)

P: Polycystic ovarian morphology (visible on ultrasound)

In 2006, the AES (Androgen Excess Society) established hyperandrogenism as the central criterion for PCOS diagnosis. The presence of multiple classification systems resulted in clinical confusion and was viewed as delaying scientific progress in our understanding of PCOS. Thus, in 2012, experts met again, and Rotterdam became the most recommended and remains the most widely used in PCOS diagnosis[3].

The evolution of these diagnostic criteria shows that PCOS is a heterogeneous disorder whose identification depends on different combinations of clinical, biochemical, and ultrasound-based evidence. The review by Christ and Cedars [3] is useful for explaining why diagnostic definitions differ and why PCOS datasets may contain different types of diagnostic information. Because predictive models are developed on datasets that represent these diagnostic decisions, this variability is crucial for machine learning research. However, as a clinical review, it does not evaluate how these types of information contribute to computational prediction.

The 2023 international evidence-based guideline further supports the Rotterdam-based view of PCOS diagnosis [1]. The guideline emphasizes that PCOS assessment involves multiple diagnostic and clinical information sources, including symptoms of ovulatory dysfunction, clinical or biochemical hyperandrogenism, and ovarian morphology. It also presents PCOS as a lifelong condition affecting the whole-body, with characteristic that may include reproductive, metabolic, dermatological, psychological, and pregnancy-related aspects. This is important for our research because it supports the idea that PCOS-related variables can be grouped according to the type of diagnostic information they represent, rather than being treated only as an unstructured list of predictors. At the same time, clinical guidelines are designed to support diagnosis and patient care, not to determine which information source is most predictive in a machine learning setting. Therefore, while diagnostic guidelines provide the clinical basis for constructing feature groups, computational studies are needed to evaluate how useful these groups are for PCOS classification. This motivates the following sections, which review machine learning approaches, feature selection methods, and model optimization techniques for PCOS prediction.

2.2 Machine Learning Approaches for PCOS Prediction

Due to the increasing accessibility of clinical datasets and improvements in computational methods, machine learning approaches have been widely applied to aid in the prediction and diagnosis of PCOS. Recent research commonly uses publicly obtainable datasets, such as the Kaggle PCOS dataset, which includes clinical, hormonal, and

[1]Monash University, *International Evidence-Based Guideline for the Assessment and Management of Polycystic Ovary Syndrome* (2023), available at: `https://www.monash.edu/medicine/mchri/pcos/guideline`

ultrasound-related features [12].

A related example is the study by Panjwani et al. [11], which proposed an optimized machine learning approach for the early detection of PCOS using initial symptoms and health indicators. The authors merged two datasets and constructed a new symptomatic dataset with 12 attributes. The selected features were grouped into broader health-related categories, including cardiovascular disease, diabetes, hypertension, obesity, and noticeable symptoms. Several machine learning models were evaluated, including K-nearest neighbors, Support Vector Machines, Logistic Regression, Decision Trees, Random Forest, XGBoost, and deep learning. The best performance was achieved by a stacked ensemble model optimized using the Walrus Optimization Algorithm, reaching an accuracy of 92.8%, AUC of 0.93, precision of 92.7%, sensitivity of 92.8%, and specificity of 92.8%. This study is useful because it shows that PCOS prediction can be approached using accessible symptom level and health indicator data, which is relevant to early detection. However, although the study groups variables into broader health categories, these groups are not compared to determine which type of information contributes most to prediction. Therefore, it supports the relevance of patient-observable and general health information, but it does not address the feature group comparison that is central to this thesis.

A different approach was proposed by Zad et al. [16], who developed machine learning models for PCOS prediction using electronic health record data. Rather than predicting only a single PCOS yes/no label, the study defined several PCOS related outcome groups, including diagnosed PCOS, irregular menstruation with hyperandrogenism, algorithmic PCOS based on Rotterdam-style criteria, and a combined diagnosed or algorithmic PCOS group. This approach is valuable because it reflects the complexity of identifying PCOS in real clinical data, where many patients may meet diagnostic criteria without having a formal diagnosis recorded. The study also shows that EHR based models may support earlier detection within healthcare systems. However, the authors note several limitations. The models were trained on an urban safety-net hospital population and still require external validation. In addition, EHR data are affected by documentation and healthcare access bias: if hormone tests, ultrasound examinations, or other measurements are not performed or recorded, the model may reflect patterns of clinical documentation rather than only the nature of the disease. This limitation is important for the present thesis because it shows that the availability of diagnostic information itself influences prediction. While Zad et al. support the relevance of hormonal and clinically observable variables, their study does not compare diagnostic-information feature groups as separate stages of assessment.

Through the rest of literature we see a wide range of supervised machine learning algorithms that have been explored for PCOS prediction, including Logistic Regression

(LR), Support Vector Machines (SVM), Decision Trees (DT), Random Forest (RF), and boosting-based models [9, 7, 2]. The performance of these models is usually evaluated using standard classification metrics such as accuracy, AUC, precision, recall, and F1-score.

In medical classification tasks such as PCOS prediction, recall is often considered more critical than accuracy, as correctly identifying positive cases is more important than minimizing false positives [6].

Previous studies indicate that ensemble learning methods, especially those based on decision trees, often achieve the best results. Random Forest and Gradient Boosting are frequently highlighted because they can model non-linear relationships and interactions between features effectively [9, 7]. In addition, some studies report very high accuracy values, reaching up to 100%, particularly when advanced ensemble methods such as stacking and XGBoost are applied [7, 2]. However, such results should be interpreted cautiously, especially when models are evaluated on relatively small datasets or limited validation procedures, as they may indicate potential overfitting.

Overall, these studies show that machine learning can support PCOS prediction using different types of data, including symptom-based information, electronic health records, clinical variables, biochemical markers, and ultrasound-derived measurements. However, much of this work focuses on overall prediction performance and algorithm comparison. Although these studies show that PCOS can be predicted using machine learning, they provide less insight into which types of diagnostic information are most responsible for model performance. This leads to the need for feature selection, explainability, and feature-group analysis, which are discussed in the following section.

2.3 Feature Selection and Model Optimization

Aside from model selection, substantial research has focused on dataset quality, feature selection, explainability, and model optimization techniques in PCOS prediction tasks. Existing literature highlights that commonly used datasets require preprocessing operations such as handling missing values, deleting redundant variables, encoding categorical features, and balancing class distributions. In one study based on the widely used Kaggle PCOS dataset, variables with a high proportion of missing values were removed, remaining missing entries were assigned using central tendency measures, categorical variables were label encoded, and class imbalance was addressed using the SMOTE-ENN resampling technique to improve classifier learning [4].

Feature selection is regularly recognized as a critical stage in enhancing prediction accuracy and interpretability. Elmannai et al. compared filter, wrapper, and embedded approaches, including mutual information, recursive feature elimination (RFE), and tree-based selection. They found that reduced feature subsets improved the performance of several machine learning models, with stacking ensembles achieving the best results [4]. This demonstrates that removing irrelevant or redundant predictors can improve PCOS classification. However, the focus remains mainly on selecting individual variables rather than evaluating broader clinically meaningful groups of diagnostic information.

A closely related study was conducted by Agirsoy and Oehlschlaeger [1], who proposed a non-invasive machine learning approach for PCOS diagnosis using ultrasound and clinical features. Their work is especially relevant because it uses the Kaggle PCOS dataset and organizes variables into clinically meaningful groups. The authors evaluated several models, including Artificial Neural Networks, Support Vector Machines, Logistic Regression, K-nearest neighbors, and XGBoost. They also applied chi-square feature selection, XGBoost feature importance, and SHAP explainability analysis to identify influential predictors. Their results showed that follicle counts, AMH, skin darkening, hair growth, weight gain, and menstrual irregularity were among the most important variables. In addition, the combination of ultrasound and clinical features achieved the strongest performance, suggesting that gynecological and clinically observable information are particularly useful for PCOS classification.

However, although this study is close to the present thesis, its feature grouping is broader and mainly follows clinical, biochemical, and ultrasound categories. These groups are useful because they align with diagnostic criteria, but they do not fully represent diagnostic information as a staged process in which some variables are easier to obtain than others. In addition, the study uses feature groups mainly to improve prediction and identify the strongest feature combination, while the present thesis focuses more directly on comparing the predictive contribution of different diagnostic-information stages. Therefore, this thesis extends this direction by separating the available predictors into patient-observable, doctor-observable, gynecological, hormonal, and blood-test feature sets and evaluating them through both direct feature-set comparison and ablation analysis.

Li et al. [8] provide a broader perspective on this issue by reviewing intelligent PCOS detection research, available datasets, and diagnostic tools. Their study is not a single prediction model, but a taxonomy and review paper. The authors argue that intelligent PCOS detection research suffers from three main problems: the lack of a standardized feature taxonomy, limited and inconsistent public datasets, and detection tools that are not sufficiently clinically validated. To address this, they proposed a taxonomy of PCOS-related features and reviewed publicly available datasets and intelligent detection tools. Their findings show that existing datasets cover only part of the proposed feature space and often lack multimodal information, while many detection tools have limitations such as high computational requirements, weak longitudinal analysis, poor multimodal processing, and limited clinical validation.

This taxonomy-based work is important for the present thesis because it supports the idea that PCOS-related variables should not be treated only as an unstructured list of predictors. Instead, they can be organized into meaningful categories of diagnostic information. However, Li et al. do not experimentally compare which feature category provides the strongest predictive performance. Their work identifies the need for clearer feature organization, while the present thesis evaluates this issue experimentally by comparing clinically motivated feature groups in PCOS classification.

Overall, these studies demonstrate that preprocessing, feature selection, explainability, and feature organization are important for PCOS classification. However, most existing research concentrates on choosing specific features, improving model performance, or proposing taxonomies of relevant variables. Less attention is given to systematically evaluating clinically significant feature groups such as patient-observable, doctor-observable, hormonal, blood-test, and gynecological variables. This creates a significant gap: it remains unclear which kinds of medical data are most predictive of PCOS and whether feature groups that are easier to obtain or less intrusive can perform comparably to specialized diagnostic features. This gap motivates the feature-group comparison and ablation design used in this thesis.

3 Methodology

In order to address the research question, we designed a machine learning experiment to compare the predictive value of different diagnostic-information feature groups for PCOS classification. Specifically, we evaluated hormonal, blood-test, patient-observable, doctor-observable, and gynecological feature groups, as well as the complete feature set containing all available predictors.

The experiment was designed to assess the contribution of these feature groups in two complementary ways. First, models were trained on individual and incremental feature sets to determine which types of diagnostic information achieve the strongest classification performance. Second, an ablation study was performed by removing one feature group at a time from the complete feature set, allowing us to examine how much performance depends on each group.

Three supervised machine learning models were used in the experiments: Logistic Regression (LR), Random Forest (RF), and Artificial Neural Network (ANN). The following sections describe the dataset, preprocessing steps, feature group construction, model training, hyperparameter optimization, validation strategy, and evaluation metrics.

All experiments were implemented in Python using the Jupyter Notebook environment, employing standard libraries such as *scikit-learn*, *NumPy*, and *Pandas*.

3.1 Dataset

This study uses the publicly available PCOS dataset from Kaggle which contains records of women evaluated for PCOS [1].

Each instance includes the following information:

- Demographic information, such as age, height, and weight ...
- Medical history
- Blood tests, including data regarding hormone, glucose, and cholesterol levels
- Physical characteristics, such as waist to hip ratio

[1] Prasad Bobby, *PCOS Data*, Kaggle dataset, available at: `https://www.kaggle.com/datasets/prasadbobby/pcosdata`

- Ultrasound examination findings to ovarian morphology

The target variable, *PCOS (Y/N)*, is binary, with 1 indicating presence and 0 indicating absence of PCOS.

Although moderate in size (541 instances), the dataset meets the typical rule of thumb for machine learning datasets, which requires at least ten observations per feature. With 41 predictor variables, the available sample size is adequate for the tests carried out in this study. The dataset is also moderately imbalanced, which is typical in medical classification problems. As demonstrated in Fig. 1, 67.28% of examples belong to the non-PCOS class, while 32.72% are PCOS cases.

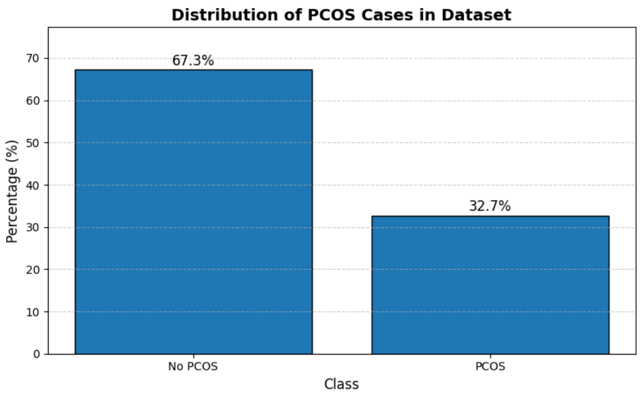

Figure 1: Class distribution of the PCOS dataset

3.2 Data Preprocessing

Data preprocessing was carried out to to standardize the dataset and prepare it for machine learning analysis. First, column names were standardized to maintain uniform formatting across the dataset. Columns containing only missing values were removed, followed by the removal of identifier-related features such as *Sl. No* and *Patient File No.*, as they do not contribute to classification.

Additionally, the characteristic *Marriage Status (Yrs)* was eliminated. Preliminary analysis indicated that specific models, notably Random Forest, gave significant importance to it despite its lack of clinical relevance for PCOS diagnosis. After removing columns, only two features remained with missing values: *AMH(ng/mL)* and *Fast food (Y/N)*, each with one missing instance. To avoid unnecessary data loss, especially when working with small dataset, no rows were eliminated at this step. Instead, missing values were handled following feature subset selection, ensuring that only relevant rows were deleted based on the feature set chosen. Categorical encoding was used for the *Cycle(R/I)* feature, which represents menstrual cycle regularity. The values were translated to binary format, with regular cycles mapped to 0 and irregular cycles mapped to 1. This transformation enables the models to treat the feature numerically instead of categorically. Following feature selection, rows with missing values in the selected features or the target variable were eliminated. This strategy minimized data loss while preserving dataset integrity.

Finally, feature scaling was implemented through standardization. To normalize feature distributions and improve model performance, a StandardScaler was fitted to the training data and used uniformly across the training, validation, and test sets. Following preprocessing, feature groups were constructed for the further analysis.

3.3 Feature Group Construction

To investigate the predictive contribution of different types of diagnostic information, feature sets were constructed according to the stage at which information may become available during the PCOS diagnostic process. The feature sets were purposely designed to be incremental rather than independent. This resembles the clinical setting in which a medical professional or gynecologist has access not only to specialized measurements but also to information obtained from the patient's history and basic clinical examination.

The feature sets, therefore, represent increasing levels of diagnostic information, from patient reported and patient-observable variables to physical examination measurements, ultrasound derived gynecological variables, laboratory blood-test variables, and the complete set of available predictors.

The following feature sets were defined:

- **Hormonal features set:** endocrine markers related to PCOS diagnosis, including FSH, LH, FSH/LH ratio, AMH, TSH, PRL, and PRG.
- **Blood-test features set:** laboratory based biomarkers, including the hormonal feature set together with additional blood test variables such as hemoglobin, vitamin D, and random blood sugar.
- **Patient-observable features set:** information that can be reported or observed without clinical testing, including age, weight, height, pregnancy history, number of abortions, menstrual cycle characteristics, weight gain, hair growth, skin darkening, hair loss, pimples, fast food consumption, and regular exercise.
- **Doctor-observable features set:** the patient-observable feature set extended with clinically measured physical parameters, including BMI, pulse rate, respiratory rate, hip and waist measurements, waist-to-hip ratio, and blood pressure.
- **Gynecologist features set:** the doctor-observable feature set is extended with ultrasound-derived gynecological measurements, including follicle count, average

follicle size, and endometrial thickness.

- **All features:** the complete set of available predictors.

This incremental grouping strategy allows the experiment to evaluate how predictive performance changes as more detailed diagnostic information becomes available. It also allows comparison between information that is easier to obtain, such as patient-observable variables, and information that requires clinical testing or specialized examination, such as hormone measurements and ultrasound derived gynecological variables.

3.3.1 Ablation study

In addition to evaluating the incremental feature sets directly, an ablation study was conducted to examine how model performance changes when a predefined feature set is excluded from the complete set. In each ablation set, one feature set was removed from the all-features set, and the remaining variables were used for model training and evaluation.

Because several feature sets are incremental, the ablation results should be interpreted as the effect of removing the corresponding diagnostic information stage as defined in this study. For example, removing the gynecological feature set removes the doctor-observable and patient-observable variables included in that incremental set, in addition to ultrasound-derived measurements. Therefore, the ablation analysis evaluates the contribution of each predefined feature set as a whole, rather than the isolated contribution of only the newly added variables.

- **All minus hormonal feature set:** all available predictors except the hormonal feature set.
- **All minus blood-test feature set:** all available predictors except the blood-test feature set.
- **All minus patient-observable feature set:** all available predictors except the patient-observable feature set.
- **All minus doctor-observable feature set:** all available predictors except the doctor-observable feature set.
- **All minus gynecological feature set:** all available predictors except the gynecological feature set.

The ablation study complements the direct comparison of incremental feature sets. The direct comparison evaluates how well each diagnostic information stage performs when used as the model input. The ablation study evaluates how performance changes when that stage is excluded from the complete predictor set. A decrease in performance after removing a feature set suggests that the removed set contributes useful predictive information. If performance remains similar or improves after removal, this suggests that the removed set may be redundant, noisy, or less informative in the context of the available dataset.

3.4 Machine Learning Models

The three models cover a simple linear method, a tree-based ensemble method, and a neural network method. This allows the experiment to test whether feature-group performance is stable across different model families.

3.4.1 Logistic Regression

Logistic Regression (LR) is a supervised machine learning algorithm used for binary classification problems, where the target variable has two possible outcomes [17]. The model operates by computing a linear combination of the input features:

$$z = wX + b$$

where X denotes the input feature vector, w denotes the model weights, and b is the bias term. The training process begins by assigning initial values to the weights and bias term, typically random or zero values. During training, these parameters are iteratively updated to minimize classification error, allowing the model to learn the relationship between the input features and the target class.

The computed value is then passed through the sigmoid function, which transforms the linear output into a probability value between 0 and 1:

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

The sigmoid function forms an "S"-shaped curve, commonly referred to as the logistic curve, making it suitable for probabilistic binary classification. In LR, a threshold value, by default 0.5, is used to determine the predicted class. If the predicted probability is equal to or greater than the threshold, the instance is classified as Class 1; otherwise, it is classified as Class 0. Logistic Regression was implemented as the baseline model for PCOS classification, providing an interpretable reference point for comparison with more complex nonlinear approaches.

3.4.2 Random Forest

Following linear Logistic Regression, Random Forest (RF) was employed as another supervised machine learning algorithm used for classification problems. Complementary to LR, RF is an ensemble learning method that combines multiple decision tree classifiers to improve predictive accuracy and control over-fitting [17]. A decision tree is a supervised learning algorithm represented as a hierarchical tree structure consisting of a root node, internal decision nodes, and leaf nodes connected through branches. At each internal node, the data are split according to a selected feature, creating increasingly homogeneous subsets. The structure resembles a flowchart in which each branch represents a possible decision path, making decision trees intuitive and easy to interpret. The model operates by training a number of decision trees on different sub-samples of the dataset. Each tree produces an individual prediction for the input feature vector X. The prediction of the b-th decision tree can be represented as:

$$T_b(X)$$

where T_b denotes the b-th tree in the forest. The final classification is then concluded by majority voting across all trees:

$$\hat{y} = \mathrm{mode}\{T_1(X), T_2(X), \ldots, T_B(X)\}$$

where B represents the total number of trees in the forest. Random Forest can capture nonlinear correlations and interactions between features, making it suitable for datasets where the relationships between variables are complex. The model also provides feature importance estimates, making it useful for feature-group analysis. Random Forest was implemented as a comparison model for PCOS classification, providing a robust nonlinear reference point for comparison with the baseline Logistic Regression model.

3.4.3 Artificial Neural Network

To complete the group, a feedforward Artificial Neural Network (ANN) was employed as the final supervised machine learning algorithm for PCOS classification. Complementary to LR and RF, ANN is a nonlinear learning model inspired by the structure and function of biological neural networks. The model consists of interconnected layers of artificial neurons, including an input layer, one or more hidden layers, and an output layer [17].

The input layer is responsible for receiving the feature values from the dataset. These values are then passed to one or more hidden layers, where the network learns patterns and relationships by transforming the information into increasingly abstract representations. Finally, the output layer produces the prediction, representing the network's final decision for the classification task.

Each neuron receives input values, applies weights and a bias term, and passes the computed value through an activation function. For a neuron in the network, the linear combination of the input features can be represented as:

$$z = wX + b$$

where X denotes the input feature vector, w denotes the weights, and b denotes the bias term. The computed value is then passed through an activation function:

$$a = f(z)$$

where $f(z)$ represents the activation function and a denotes the activated output of the neuron. During training, the network adjusts its weights and biases through a process called backpropagation, which aims to reduce prediction error across the training data. In the hidden layers, nonlinear activation functions allow the network to learn complex patterns and relationships between features. For binary classification, the final output layer commonly uses the sigmoid activation function to produce a probability value between 0 and 1:

$$\hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}}$$

A threshold value, by default 0.5, is then used to determine the predicted class. If the predicted probability is equal to or greater than the threshold, the instance is classified as Class 1; otherwise, it is classified as Class 0. ANN can capture complex nonlinear correlations and interactions between input features, making it suitable for structured medical datasets where relationships between variables may not be linear. In this experiment, ANN was implemented as the deep learning model for PCOS classification, allowing for a comparison between classical machine learning methodologies and neural network-based approaches.

3.5 Hyperparameter Optimization

Each classification algorithm had its hyperparameters optimized to increase model performance. Given the medical context of PCOS diagnosis, recall was chosen as the major optimization parameter for reducing false negatives.

3.5.1 Logistic Regression

Logistic Regression hyperparameters were tuned using a grid search technique.

- The regularization strength parameter C : $\{0.1, 1, 2\}$
- Class weight: class weights were changed to resolve class imbalance and emphasize the proper classification of PCOS cases. The positive class (PCOS = 1) was assigned larger weights, with configurations $\{1 : 1\}$, $\{1 : 1.5\}$, and $\{1 : 2\}$.

3.5.2 Random Forest

For the Random Forest model, a grid search was performed over multiple hyperparameters, including:

- Number of trees (*n_estimators*): $\{100, 200, 300\}$
- Maximum tree depth (*max_depth*): $\{\text{None}, 5, 10, 15\}$
- Number of features considered at each split (*max_features*): $\{\text{sqrt}, \log 2\}$
- Minimum samples per leaf (*min_samples_leaf*): $\{1, 2, 4\}$
- Class weights: $\{1 : 1\}$, $\{1 : 1.5\}$, $\{1 : 2\}$

Similarly to Logistic Regression, class weighting was applied to prioritize recall and reduce false negatives.

3.5.3 Artificial Neural Network

The artificial neural network's hyperparameters were manually tuned using a grid search-like approach. The following parameters were analyzed:

- Number of neurons per hidden layer: $\{8, 16, 32\}$
- Dropout rate: $\{0, 0.2\}$
- Learning rate: $\{0.005, 0.001\}$
- Batch size: $\{16, 32, 64\}$
- Number of epochs: 65

The network architecture consisted of two fully linked hidden layers with ReLU activation, followed by a sigmoid output layer for binary classification. Dropout regularization was used to minimize overfitting.

Class weighting was also included in the training process, with the positive class (PCOS = 1) receiving a higher weight to improve recall.

The best-performing model was chosen based on validation recall to ensure accurate identification of PCOS cases.

3.6 Validation Strategy

To evaluate model performance and reduce the risk of overfitting, the dataset was divided into training, validation, and test subsets. The training and validation data were used during model development and hyperparameter selection, while the test set was kept separate and used only for the final evaluation. This ensured that the final reported results were obtained on unseen data.

For Logistic Regression and Random Forest, hyperparameters were selected using grid search. Different parameter combinations were tested, and the best configuration was chosen based on recall, since the main goal was to correctly identify PCOS-positive cases and reduce false negatives. After selecting the best hyperparameters, the final model was trained and evaluated on the held-out test set.

Cross-validation was also used as an additional check of model stability and generalization. The cross-validation recall scores were not used as the final test results, but rather as an indicator of whether the model performance was consistent across different data splits. Large differences between cross-validation performance and test performance would suggest possible overfitting or instability.

For the Artificial Neural Network, hyperparameters were selected using validation-set performance rather than cross-validation, because repeated neural network training is computationally more expensive and less stable. The best ANN configuration was chosen based on validation recall and then evaluated on the same held-out test set as the other models.

The final comparison between models and feature sets was therefore based on test-set performance, while cross-validation scores were used as supporting evidence for model robustness.

3.7 Evaluation Metrics

Model performance was examined using various standard classification metrics:

- **Accuracy:** The percentage of successfully classified instances among all predictions.
- **Precision:** The proportion of true positive predictions among all predicted positive cases.
- **Recall:** Measures the proportion of actual positive cases that were correctly identified by the model.
- **F1-score:** The harmonic mean of precision and recall, providing a balance between the two.

- **Confusion Matrix:** Provides a detailed breakdown of prediction outcomes, including True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN).

Recall was treated as the main optimization metric because false negatives are especially important in PCOS classification. A false negative occurs when a patient with PCOS is incorrectly classified as non-PCOS, which may delay further diagnosis or treatment. Therefore, models were tuned to prioritize recall, even when this reduced precision. The final model's performance on the held-out test set was evaluated using all metrics, and nested cross-validation recall scores were used only as supporting evidence of model stability across different data splits.

Furthermore, the results were examined across feature groups to determine the impact of various forms of clinical information on prediction performance.

4 Results

This chapter presents the results of the PCOS classification experiments in relation to the research question: which diagnostic information feature groups provide the strongest predictive performance for PCOS classification. The results are organized around the comparison of feature groups rather than around individual models because the main goal of this thesis is to evaluate the predictive value of different types of diagnostic information.

First, the performance of incremental feature sets is reported to examine how classification performance changes as additional groups of features are included. Second, the ablation analysis is presented to show how model performance changes when each feature group is removed from the complete feature set. Third, the performance of the three machine learning models is compared across the evaluated feature sets. Finally, feature importance and false positive/false negative analyzes are used to further interpret which variables and feature groups contributed most strongly to PCOS classification.

Model performance is evaluated using accuracy, precision, recall, F1-score, and false positive and false negative counts. Particular attention is given to recall and false negatives because false negative predictions correspond to PCOS cases that are incorrectly classified as non-PCOS, which may delay further diagnosis or treatment. Complete performance results for all models and feature sets are provided in Appendix A.

4.1 Performance of Incremental Feature Sets

Table 2 summarizes the strongest performing model for each incremental feature set. The model shown for each feature set was selected based on the overall balance between accuracy, recall, F1-score, and false negative count.

Table 2: Strongest performing model for each incremental feature set.

Feature set	Model	Acc	Rec	Prec	F1	CV Rec
Hormonal	RF	0.689	0.622	0.528	0.571	0.551
Blood-test	RF	0.667	0.622	0.500	0.555	0.533
Patient-observable	LR	0.815	0.804	0.698	0.748	0.771
Doctor-observable	ANN	0.807	0.761	0.700	0.729	–
Gynecological	RF	0.919	0.957	0.830	0.889	0.721
All features	RF	0.889	0.854	0.796	0.824	0.789

The choice of feature groups significantly affected the classification performance of every model that was assessed. This directly answers the study question of what kinds of clinical data are most important for predicting PCOS.

Overall, the biochemical feature sets, represented by the *Hormonal* and *Blood-test* feature set, produced the weakest results among the incremental feature sets. The *Hormonal* feature set achieved an F1-score of 0.571 with 17 false negatives, while the *Blood-test* feature set achieved an F1-score of 0.555 with the same number of false negatives. In contrast, the *Patient-observable* feature set performed substantially better, reaching an accuracy of 0.815, a recall of 0.804, and F1-score of 0.748. This suggests that symptoms and characteristics that can be reported or observed without clinical testing contain useful predictive information. The strongest results were obtained by the *Gynecological* and *All features* feature sets. Although the *All features* set performed strongly, with an accuracy of 0.889, a recall of 0.854, and a F1-score of 0.824, it was outperformed by the *Gynecological* feature set, which achieved the highest accuracy of 0.919, a recall of 0.957, F1-score of 0.889, and only two false negatives.

4.2 Ablation Analysis

Table 3 summarizes the strongest performing model for each ablation feature set. In each ablation feature set, one predefined feature subset was removed from the complete feature set. Because the feature sets are incremental, the ablation results refer to the removal of the predefined feature set as a whole.

Table 3: Strongest performing model for each ablation feature set.

Removed feature set	Model	Acc	Rec	Prec	F1	CV Rec
Hormonal	RF	0.926	0.957	0.846	0.898	0.741
Blood-test	RF	0.919	0.957	0.830	0.889	0.761
Patient-observable	RF	0.844	0.778	0.761	0.769	0.616
Doctor-observable	RF	0.807	0.733	0.702	0.717	0.643
Gynecological	RF	0.659	0.578	0.491	0.531	0.542

The ablation results further confirm that gynecological information contributed most strongly to PCOS classification. Removing the *Hormonal* feature set produced the strongest ablation result, with an accuracy of 0.926, recall of 0.957, precision of 0.846, F1-score of 0.898, and only two false negatives. Removing the *Blood-test* feature set produced a very similar result, with an accuracy of 0.919, recall of 0.957, F1-score of 0.889, and two false negatives. This suggests that, in this dataset, hormonal and blood-test variables did not add substantial predictive information when the remaining clinical and gynecological variables were available.

In contrast, removing the *Gynecological* feature set caused the largest performance decline. In this ablation feature set, Random Forest achieved only 0.659 accuracy, 0.578 recall, and an F1-score of 0.531, with 19 false negatives. Removing the *Patient-observable* and *Doctor-observable* feature sets also reduced performance, producing F1-scores of 0.769 and 0.717, respectively. These results indicate that the model relied most heavily on gynecological information, while patient-observable and doctor-observable variables also provided useful predictive information.

4.3 Comparison of Machine Learning Models

Although the main focus of this thesis is the comparison of feature groups, model performance also differed across algorithms. Overall, Random Forest provided the strongest balance across accuracy, recall, precision, F1-score, and false negative count. It achieved the best result for the *Gynecological* incremental feature set and the strongest ablation result when the *Hormonal* feature set was removed.

Logistic Regression remained a useful, interpretable baseline and performed competitively in some feature sets, especially for the *Patient-observable* feature set. However, its performance was less balanced across all metrics than Random Forest.

The Artificial Neural Network often achieved high recall, but this was accompanied by many false positives and lower precision, particularly for the *Hormonal* and *Blood-test* feature sets. This may be related to the moderate dataset size and class imbalance, since neural networks are often more sensitive to unstable or imbalanced training data. Therefore, ANN was less reliable than Random Forest in this experiment.

4.4 Feature Importance Analysis

To further examine the contribution of individual variables, Random Forest feature importance analysis was performed for all evaluated feature sets. In this section, only the three most relevant sets are shown: *All features*, *All minus hormones*, and *Gynecological*. These sets were selected because they achieved the strongest overall performance.

The remaining feature importance visualizations are provided in Appendix B. Because feature importance values were obtained from Random Forest models, they reflect the behavior of this model family and should not be generalized to all classifiers.

Across the three selected sets, gynecological variables were consistently ranked as the most important predictors. In the *All features* set, the two highest ranked variables were follicle count in the right ovary and follicle count in the left ovary, both of which describe ovarian morphology. These were followed by several patient-observable symptoms, especially hair growth and menstrual cycle irregularity. Only two biochemical variables appeared among the highest ranked predictors: AMH and the FSH/LH ratio.

A similar pattern was observed in the *All minus hormones* set. Even after removing hormonal variables, follicle counts in both ovaries remained the strongest predictors, followed by patient-observable symptoms. BMI also appeared among the important variables in this set, suggesting that doctor-observable physical measurements contributed additional information when hormonal markers were excluded.

The *Gynecological* feature set showed the same overall pattern. The strongest predictors were again related to ovarian morphology, while the remaining important variables were mainly patient-observable or clinically observable symptoms. This supports the main result that gynecological information contributed most strongly to PCOS classification, while patient-observable symptoms also provided useful predictive information.

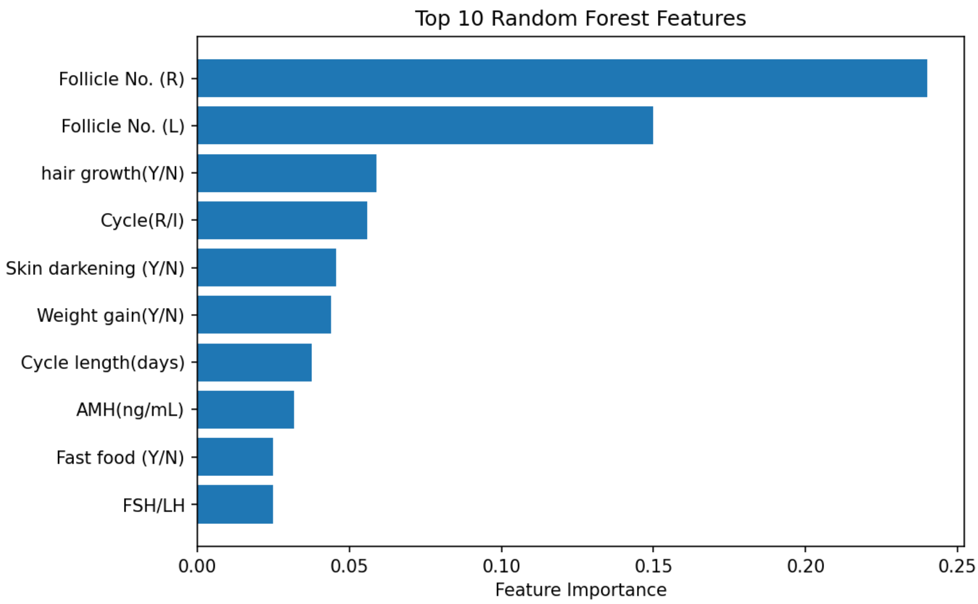

Figure 2: Feature importance scores for the all_features set obtained using Random Forest.

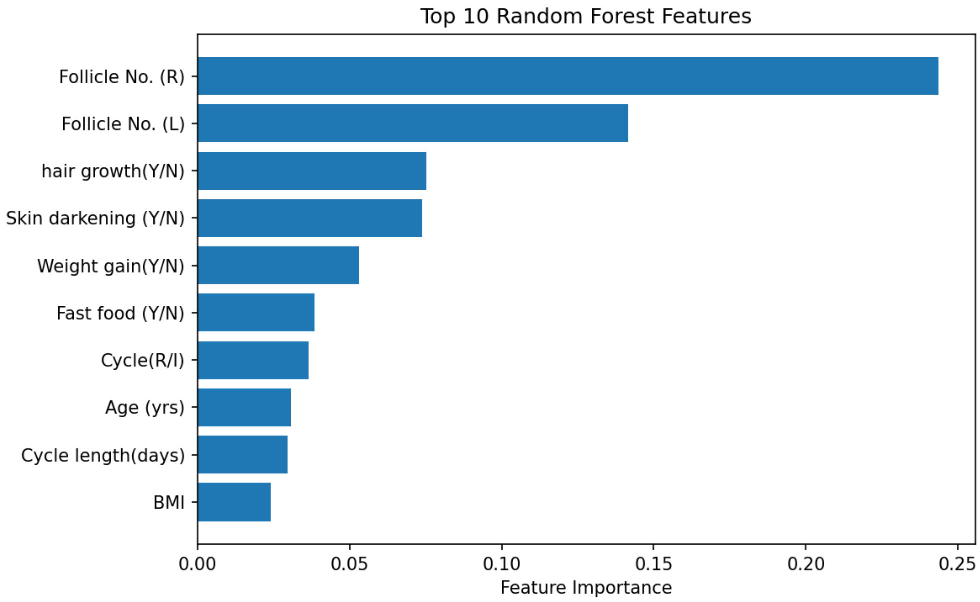

Figure 3: Feature importance scores for the all_minus_hormones set obtained using Random Forest.

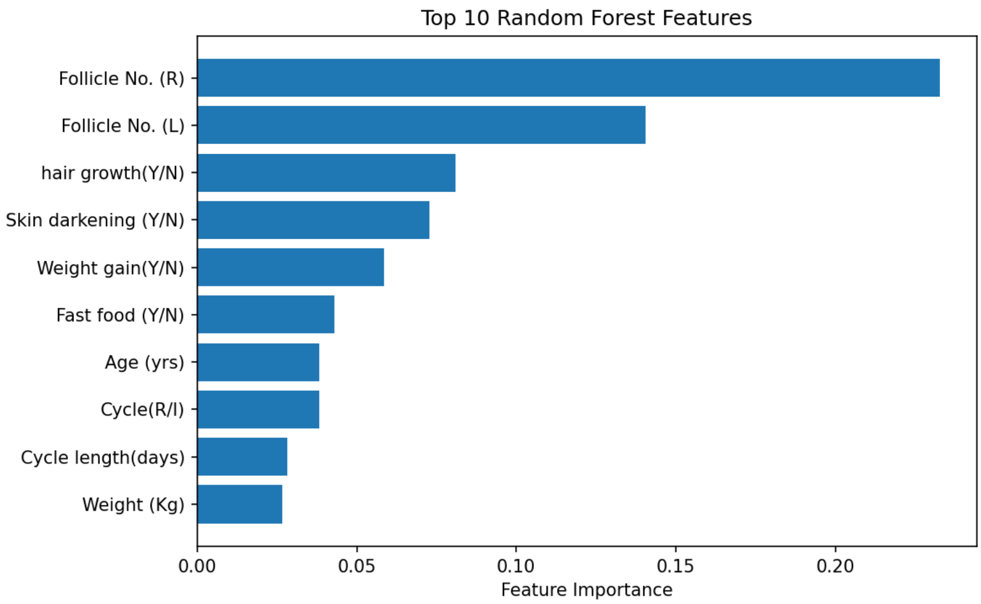

Figure 4: Feature importance scores for the gynecological feature set obtained using Random Forest.

4.5 False Positive and False Negative Analysis

False positive and false negative counts were examined to compare the types of classification errors made by the models. False negatives are especially relevant in this study because they represent PCOS positive cases that were incorrectly classified as non-PCOS.

The lowest false negative counts were obtained by the *Gynecological* incremental feature set and the *Hormonal removed* ablation set. In both cases, Random Forest produced only two false negatives. In contrast, removing the *Gynecological* feature set produced the largest number of false negatives for Random Forest, with 19 PCOS-positive cases incorrectly classified as non-PCOS.

The table also shows a trade-off between false negatives and false positives. ANN reduced false negatives in the *Hormonal* feature set, producing six false negatives compared with 17 for Random Forest, however this came with 72 false positives. A similar pattern appeared in the broader results for the *Blood-test* feature set. Therefore, although ANN sometimes improved recall, Random Forest provided a more balanced error profile across the most relevant sets.

4.6 Summary of Key Findings

The results show that feature-set composition had a substantial effect on PCOS classification performance. Among the incremental feature sets, the *Gynecological* feature set achieved the strongest overall result, with an accuracy of 0.919, a recall of 0.957, and a F1-score of 0.889. In contrast, the *Hormonal* and *Blood-test* feature sets performed the weakest, with F1-scores of 0.571 and 0.555, respectively. This suggests that ovarian morphology and related gynecological information had stronger predictive value than biochemical markers alone in this dataset.

The ablation analysis showed a similar pattern. Removing the *Hormonal* or *Blood-test* feature sets did not reduce model performance. The strongest overall ablation feature set was obtained when the *Hormonal* feature set was removed, where Random Forest achieved an accuracy of 0.926, a recall of 0.957, and F1-score of 0.898. In contrast, removing the *Gynecological* feature set caused the largest performance decline, reducing the F1-score to 0.531 and increasing the false negative count to 19.

Across models, Random Forest achieved the strongest overall balance across most feature sets. Logistic Regression performed competitively in some feature sets, particularly the *Patient-observable* and *All features* feature sets. ANN often achieved high recall, but this came at the cost of increased false positive counts and lower precision. Therefore, Random Forest was the most stable model for comparing feature-set performance in this study.

5 Discussion and Conclusion

5.1 Interpretation from Results

The results show that feature-set composition played an important role in PCOS classification performance. Across both the incremental feature set experiments and the ablation analysis, the *Gynecological* feature set provided the strongest predictive information. This suggests that ultrasound-derived measurements, especially follicle-related variables, were the most informative predictors for PCOS classification in this dataset.

The ablation analysis further supports this interpretation. Removing the *Gynecological* feature set produced the weakest performance and the highest number of false negatives. However, this result must be interpreted carefully because the feature sets in this thesis were designed incrementally. Therefore, removing the *Gynecological* feature set also removes the *Doctor-observable* and *Patient-observable* variables included in that stage. As a result, the *All minus gynecological* feature set mainly contains biochemical information, including hormonal and blood-test variables. The poor performance of this ablation feature set, therefore suggests that biochemical markers alone were not sufficient for strong PCOS classification in this dataset.

This is an important finding because PCOS is often described as an endocrine disorder, and hormonal markers such as AMH and the FSH/LH ratio are commonly associated with PCOS diagnosis. However, in this study, the *Hormonal* and *Blood-test* feature sets performed relatively weakly when used on their own. At the same time, biochemical markers were not completely unimportant. In the Random Forest feature-importance analysis, AMH and the FSH/LH ratio appeared among the most important predictors when combined with other types of information. This suggests that biochemical markers may contribute useful information, but they were not sufficient as isolated feature sets in this experiment.

The *Patient-observable* feature set also performed well compared with the *Hormonal* and *Blood-test* feature sets. In addition, Random Forest feature-importance results showed that patient-observable symptoms frequently appeared among the most important predictors. These included skin darkening, hair growth, and menstrual cycle irregularity. This indicates that symptoms and characteristics that can be reported or noticed without clinical testing still carry meaningful predictive information.

Overall, the results suggest that each group of diagnostic information contributes some predictive value, but not to the same extent. In this dataset, gynecological information provided the strongest contribution; patient-observable and doctor-observable information also contributed useful predictive value, while biochemical markers were less effective when used alone. These findings suggest that accessible and less invasive information may help identify patients who should seek further clinical evaluation, but it should not be treated as a replacement for medical diagnosis.

5.2 Comparison with Related Work

The findings of this thesis are consistent with previous studies showing that ensemble models and ultrasound-derived features are useful for PCOS prediction. Several studies have reported strong performance from tree-based ensemble methods such as Random Forest, Gradient Boosting, stacking, and XGBoost [9, 7]. Similarly, in this thesis, Random Forest provided the most stable overall performance across feature sets, especially for the *Gynecological* feature set and the strongest ablation feature sets.

The results are also consistent with the study by Agirsoy and Oehlschlaeger [1], who found that ultrasound and clinical features produced strong PCOS classification performance. Their feature importance and explainability analyzes identified follicle counts, AMH, hair growth, weight gain, and menstrual irregularity as influential predictors. The present thesis supports these findings because follicle-count variables were consistently among the strongest predictors, and the *Gynecological* feature set achieved the best performance among the incremental feature sets.

The findings also relate to symptom-based PCOS prediction studies, such as Panjwani et al. [11], which showed that early detection can be supported using symptoms and general health indicators. In this thesis, the *Patient-observable* feature set performed substantially better than the *Hormonal* and *Blood-test* feature sets. This supports the idea that symptom-level information may be useful for early screening or preliminary risk assessment.

At the same time, this thesis differs from much of the existing literature. Many previous studies focus mainly on comparing machine learning algorithms, optimizing model performance, or identifying individual important predictors. In contrast, the main focus of this thesis was to compare clinically meaningful diagnostic information feature sets. This makes the results more directly related to the practical question of which types of information are most useful at different stages of PCOS assessment.

5.3 Clinical and Practical Implications

The results suggest that patient-observable symptoms should not be ignored. Symptoms such as menstrual irregularity, hair growth, skin darkening, and weight gain may provide useful early signals that a patient should seek further medical evaluation. Since these features are easier to notice and do not require laboratory testing or ultrasound examination, they may be useful for early awareness and preliminary screening.

However, the results also show that patient-observable information cannot replace clinical evaluation. The strongest performance was achieved by the *Gynecological* feature set, which includes ultrasound derived information together with patient-observable and doctor-observable variables. This suggests that further medical examination remains important for reliable PCOS classification. Therefore, less invasive information may help indicate when further assessment is needed, but diagnosis should still be supported by clinical, biochemical, and/or gynecological evaluation.

These findings may also be useful when designing staged screening approaches. For example, patient-observable symptoms could be used as an early step to identify individuals who may benefit from further testing. Doctor-observable measurements and gynecological examination could then provide more specific diagnostic information. Such an approach may be especially useful in contexts where access to specialist testing is limited or delayed.

5.4 Limitations of the Study

This study has several limitations. First, the analysis was based on a single publicly available dataset. Although the dataset is useful for experimentation, it is moderate in size and may not fully represent the wider PCOS population. The results may, therefore, not generalize to other clinical settings, populations, or healthcare systems without external validation.

Second, the study was conducted as a bachelor-level research project by a computer science student. This means that the work was limited by the available time, resources, and level of experience. In particular, the medical, biochemical, and endocrinological interpretation of the results may be limited because the thesis was conducted from a primarily computational perspective.

Third, the feature sets were designed as incremental diagnostic-information stages. This makes the setup clinically meaningful, but it also means that some feature sets overlap. For example, the *Gynecological* feature set includes *Patient-observable* and *Doctor-observable* variables in addition to ultrasound-derived measurements. Therefore, the ablation results should be interpreted as the removal of predefined diagnostic-information stages rather than isolated independent feature groups.

Fourth, only a limited number of machine learning models were evaluated. Logistic Regression, Random Forest, and Artificial Neural Network were chosen to represent a linear model, a tree-based ensemble model, and a neural network model. However, other models, such as XGBoost, Gradient Boosting, Support Vector Machines, or more advanced deep learning architectures, may produce different results.

Finally, feature importance was evaluated using Random Forest. Therefore, the feature-importance results reflect the behavior of this model family and should not be generalized to all possible classifiers.

5.5 Future Work

Future work should validate these findings on larger and more diverse datasets. External validation would be especially important to determine whether the strong performance of the *Gynecological* feature set and the usefulness of *Patient-observable* symptoms remain stable across different populations and clinical settings.

Future research could also compare independent, non-overlapping feature groups. This would make it possible to estimate the isolated contribution of patient-observable symptoms, doctor-observable measurements, gynecological variables, hormonal markers, and blood-test variables more precisely.

Another direction would be to include a wider range of machine learning models and optimization strategies. Models such as XGBoost, Gradient Boosting, and Support Vector Machines could be compared with the models used in this thesis. In addition, explainable artificial intelligence methods such as SHAP could be applied across multiple models to better understand how different feature groups influence predictions.

Finally, future studies could benefit from interdisciplinary collaboration. Combining expertise from computer science, medicine, endocrinology, and gynecology would allow for a stronger interpretation of both the computational results and the clinical meaning of the selected features. This could support the development of more reliable and clinically useful PCOS screening tools.

5.6 Conclusion

This study investigated which diagnostic information feature sets provide the strongest predictive performance for PCOS classification. To address this question, a machine learning experiment was designed using feature groups constructed incrementally to reflect the diagnostic process. In addition to evaluating these incremental feature sets directly, an ablation study was conducted to examine how model performance changed when each diagnostic information stage was removed.

The results showed that the strongest predictive information came from ultrasound derived gynecological measurements, especially follicle count. The *Gynecological* feature set achieved the strongest overall performance, while removing this feature set caused the largest decline in classification performance. Patient-observable symptoms also played an important role, suggesting that symptoms noticed by patients may provide a useful foundation for further clinical assessment.

Overall, this experiment suggests that evaluating feature sets can help clarify which types of diagnostic information are most useful for PCOS classification. The findings may support the development of staged screening approaches, where accessible patient-observable information is used as an early signal, and more specialized clinical or gynecological information is used for further assessment. However, because this study was conducted on a relatively small public dataset, external validation on larger and more diverse clinical datasets is needed.

Povzetek

Sindrom policističnih jajčnikov (PCOS) je pogosta endokrina motnja pri ženskah v reproduktivnem obdobju. Ker se lahko kaže z različnimi kombinacijami simptomov, biokemičnih označevalcev in ultrazvočnih ugotovitev, je njegova diagnoza pogosto zahtevna. V praksi se pri ocenjevanju PCOS uporabljajo različne vrste diagnostičnih informacij, na primer podatki, ki jih lahko opazi pacientka sama, meritve, ki jih opravi zdravnik, laboratorijski izvidi, hormonske vrednosti in ginekološki oziroma ultrazvočni podatki. Namen zaključne naloge je bil zato raziskati, kateri nabori diagnostičnih informacij imajo največjo napovedno vrednost pri klasifikaciji PCOS z metodami strojnega učenja.

V raziskavi je bila uporabljena javno dostopna podatkovna zbirka PCOS, ki vsebuje demografske podatke, podatke o zdravstveni anamnezi, laboratorijske in hormonske meritve, fizične značilnosti ter ultrazvočne podatke. Ciljna spremenljivka je bila binarna in je označevala prisotnost ali odsotnost PCOS. Pred modeliranjem so bili podatki predobdelani: odstranjeni so bili identifikacijski in nerelevantni stolpci, urejena so bila manjkajoča polja, kategorične spremenljivke so bile pretvorjene v numerično obliko, izbrane značilke pa standardizirane. Značilke so bile nato razdeljene v več diagnostično smiselnih skupin: hormonski nabor, krvni nabor, nabor značilk, ki jih lahko opazi pacientka, nabor značilk, ki jih lahko opazi oziroma izmeri zdravnik, ginekološki nabor in celoten nabor značilk.

Eksperimentalni del je bil zasnovan na dveh pristopih. Najprej so bili modeli naučeni na posameznih oziroma postopno razširjenih naborih značilk, da bi se pokazalo, kako se uspešnost spreminja z dodajanjem novih vrst diagnostičnih informacij. Nato je bila izvedena ablacijska analiza, pri kateri je bil iz celotnega nabora značilk odstranjen po en vnaprej določen nabor. S tem je bilo mogoče oceniti, koliko posamezna skupina prispeva k uspešnosti klasifikacije, kadar so na voljo tudi druge informacije. Uporabljeni so bili trije modeli strojnega učenja: logistična regresija, naključni gozd in umetna nevronska mreža. Uspešnost modelov je bila ocenjena z metrikami točnost, natančnost, priklic, F1-mera ter s številom lažno pozitivnih in lažno negativnih napovedi. Posebna pozornost je bila namenjena priklicu, saj je pri medicinski klasifikaciji pomembno zmanjšati število primerov, pri katerih je bolnica s PCOS napačno

razvrščena kot oseba brez PCOS.

Rezultati so pokazali, da sestava nabora značilk pomembno vpliva na uspešnost klasifikacije. Hormonski in krvni nabor sta pri samostojni uporabi dosegla najšibkejše rezultate, kar kaže, da v uporabljeni podatkovni zbirki sama laboratorijska oziroma hormonska informacija ni bila dovolj za najbolj zanesljivo napoved. Nabor značilk, ki jih lahko opazi pacientka, je dosegel boljše rezultate, kar pomeni, da imajo simptomi in osnovni podatki, dostopni brez zahtevnejšega kliničnega testiranja, uporabno napovedno vrednost. Najboljše rezultate med postopnimi nabori je dosegel ginekološki nabor, ki je vključeval tudi ultrazvočne podatke, zlasti število foliklov in morfologijo jajčnikov. Ta nabor je dosegel visoko vrednost priklica in najmanj lažno negativnih napovedi.

Ablacijska analiza je dodatno potrdila pomen ginekoloških informacij. Odstranitev hormonskega ali krvnega nabora iz celotnega nabora značilk ni poslabšala uspešnosti modelov, v nekaterih primerih pa so rezultati ostali skoraj enaki ali se celo izboljšali. Nasprotno pa je odstranitev ginekološkega nabora povzročila največji padec uspešnosti, kar kaže, da so bile ginekološke oziroma ultrazvočne značilke najpomembnejše za napoved PCOS v tej podatkovni zbirki. Tudi odstranitev pacientu opaznih in zdravniku opaznih značilk je zmanjšala uspešnost, kar potrjuje, da imajo tudi te informacije pomembno podporno vlogo.

Med uporabljenimi modeli je naključni gozd najpogosteje dosegel najbolj uravnotežene rezultate glede na točnost, priklic, F1-mero in število lažno negativnih primerov. Logistična regresija je bila uporabna kot preprost in razložljiv osnovni model, umetna nevronska mreža pa je v nekaterih primerih dosegla dober priklic, vendar z manj stabilnim ravnovesjem med metrikami. Analiza pomembnosti značilk je pokazala, da so med najvplivnejšimi napovedniki število foliklov, nekatere ultrazvočne meritve in simptomi, ki jih lahko opazi pacientka.

Zaključna naloga ugotavlja, da različne vrste diagnostičnih informacij ne prispevajo enako h klasifikaciji PCOS. V uporabljeni podatkovni zbirki so bile najmočnejše ginekološke informacije, medtem ko so pacientu opazni simptomi prav tako zagotavljali koristne napovedne informacije. Strukturirano vrednotenje naborov značilk lahko zato izboljša razumevanje modelov strojnega učenja in pomaga določiti, katere informacije so najvrednejše za podatkovno podprto odkrivanje PCOS. V prihodnje bi bilo treba rezultate preveriti na večjih, raznolikih in klinično bolje potrjenih podatkovnih zbirkah ter dodatno raziskati uporabnost takšnih modelov v praksi.

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Appendices

A Additional Metric Evaluation Tables.

Incremental Feature Sets

Table 4 reports the complete metric results for all models evaluated on the incremental feature sets.

Table 4: Model performance across incremental feature sets.

Feature set	Model	Acc	Rec	Prec	F1	TN	FP	FN	TP	CV Rec
Hormonal	LR	0.467	0.756	0.358	0.486	29	61	11	34	0.523
Hormonal	RF	0.689	0.622	0.528	0.571	65	25	17	28	0.551
Hormonal	ANN	0.422	0.867	0.351	0.500	18	72	6	39	–
Blood-test	LR	0.430	0.822	0.349	0.490	21	69	8	37	0.505
Blood-test	RF	0.667	0.622	0.500	0.555	62	28	17	28	0.533
Blood-test	ANN	0.407	0.911	0.350	0.506	14	76	4	41	–
Patient-observable	LR	0.815	0.804	0.698	0.748	73	16	9	37	0.771
Patient-observable	RF	0.815	0.696	0.744	0.719	78	11	14	32	0.741
Patient-observable	ANN	0.807	0.783	0.692	0.735	73	16	10	36	–
Doctor-observable	LR	0.793	0.739	0.680	0.708	73	16	12	34	0.771
Doctor-observable	RF	0.815	0.696	0.744	0.719	78	11	14	32	0.701
Doctor-observable	ANN	0.807	0.761	0.700	0.729	74	15	11	35	–
Gynecological	LR	0.844	0.957	0.698	0.807	70	19	2	44	0.770
Gynecological	RF	0.919	0.957	0.830	0.889	80	9	2	44	0.721
Gynecological	ANN	0.830	0.913	0.689	0.785	70	19	4	42	–
All features	LR	0.874	0.902	0.740	0.813	81	13	4	37	0.859
All features	RF	0.889	0.854	0.796	0.824	85	9	6	35	0.789
All features	ANN	0.830	0.878	0.667	0.758	76	18	5	36	–

Ablation Study Feature Sets

Table 5 reports the complete metric results for all ablation configurations.

Table 5: Ablation study model performance with feature-set removal.

Removed feature set	Model	Acc	Rec	Prec	F1	TN	FP	FN	TP	CV Rec
Hormonal	LR	0.822	0.935	0.672	0.782	68	21	3	43	0.770
Hormonal	RF	0.926	0.957	0.846	0.898	81	8	2	44	0.721
Hormonal	ANN	0.874	0.935	0.754	0.835	75	14	3	43	–
Blood-test	LR	0.837	0.957	0.688	0.800	69	20	2	44	0.780
Blood-test	RF	0.919	0.957	0.830	0.889	80	9	2	44	0.731
Blood-test	ANN	0.844	0.935	0.705	0.804	71	18	3	43	–
Patient-observable	LR	0.807	0.800	0.679	0.735	73	17	9	36	0.855
Patient-observable	RF	0.844	0.778	0.761	0.769	79	11	10	35	0.791
Patient-observable	ANN	0.807	0.822	0.673	0.740	72	18	8	37	–
Doctor-observable	LR	0.770	0.822	0.617	0.705	67	23	8	37	0.873
Doctor-observable	RF	0.807	0.733	0.702	0.717	76	14	12	33	0.791
Doctor-observable	ANN	0.778	0.822	0.627	0.712	68	22	8	37	–
Gynecological	LR	0.437	0.822	0.352	0.493	22	68	8	37	0.505
Gynecological	RF	0.659	0.578	0.491	0.531	63	27	19	26	0.552
Gynecological	ANN	0.504	0.689	0.369	0.481	37	53	14	31	–

B Additional Feature Importance Visualizations.

This appendix contains additional Random Forest feature importance visualizations for feature sets that are not shown in the main Results chapter.

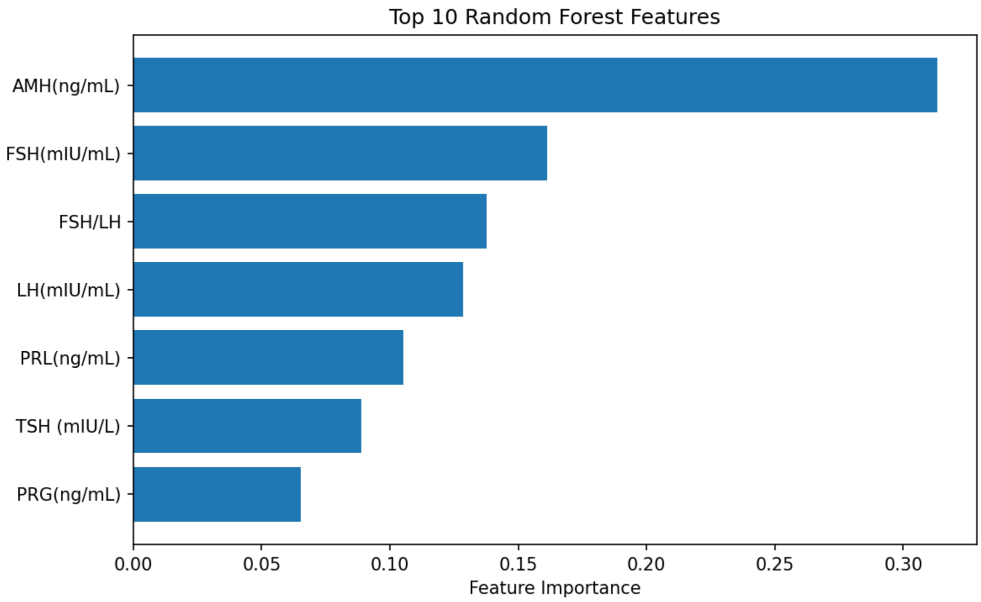

Figure 5: Feature importance scores for the hormonal feature set obtained using Random Forest.

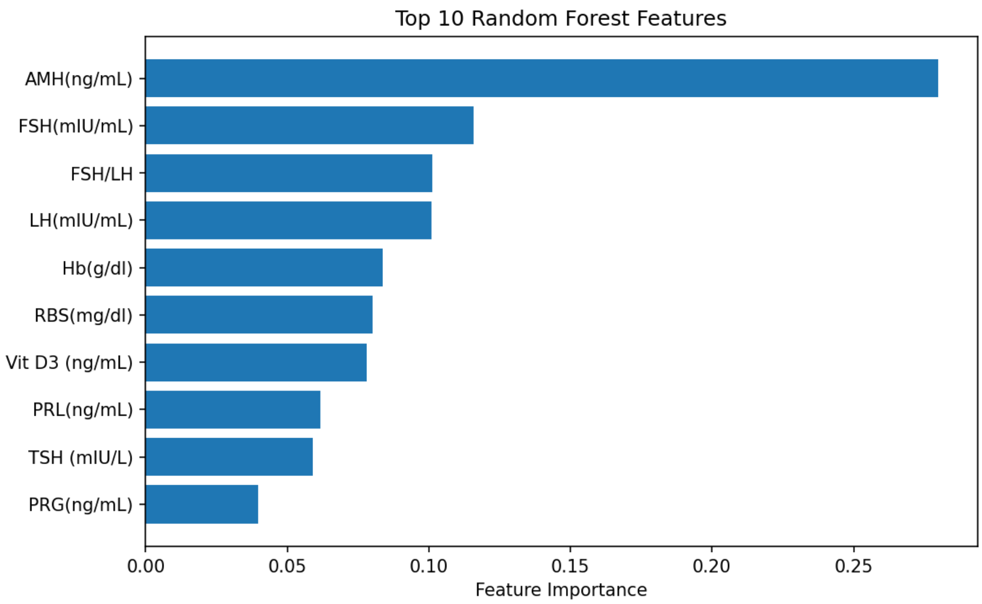

Figure 6: Feature importance scores for the blood-test feature set obtained using Random Forest.

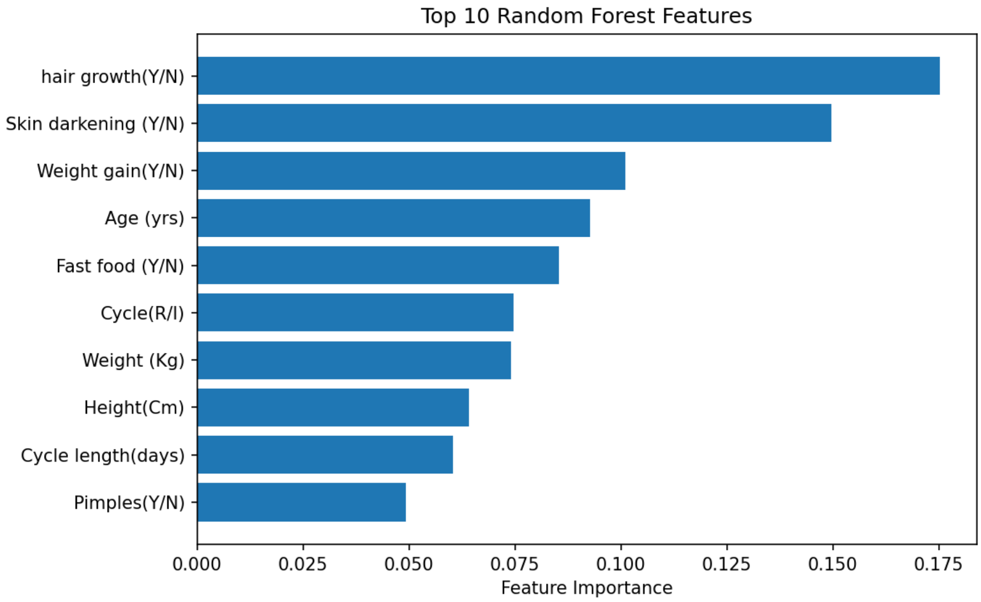

Figure 7: Feature importance scores for the patient-observable feature set obtained using Random Forest.

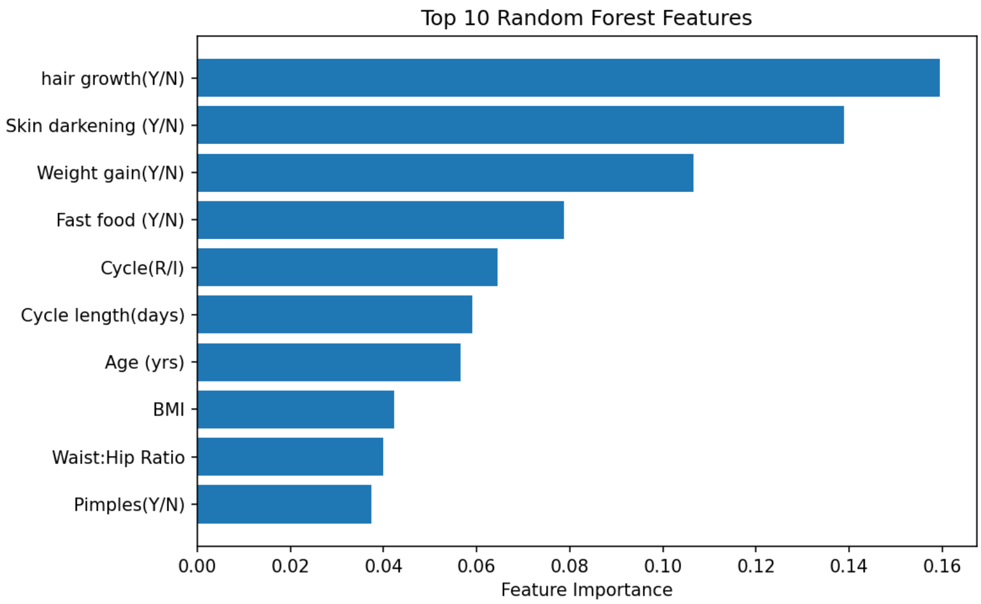

Figure 8: Feature importance scores for the doctor-observable feature set obtained using Random Forest.

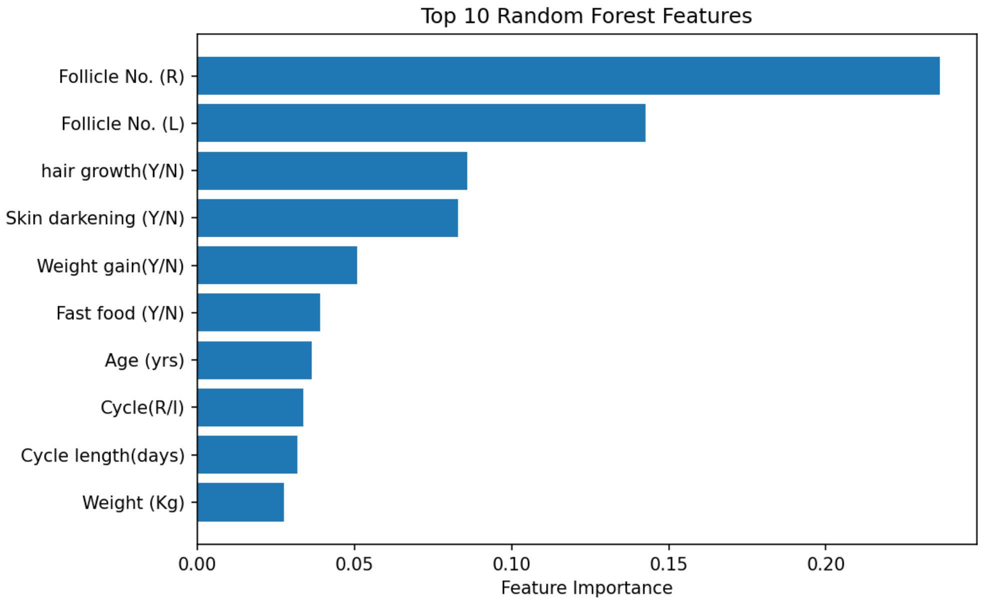

Figure 9: Feature importance scores for the all-minus-blood-test feature set obtained using Random Forest.

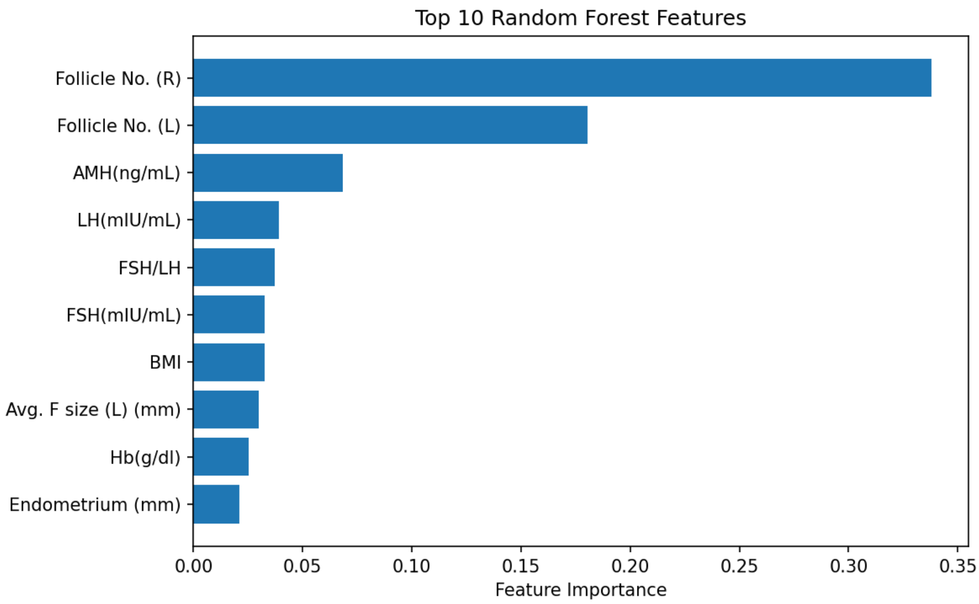

Figure 10: Feature importance scores for the all-minus-patient-observable feature set obtained using Random Forest.

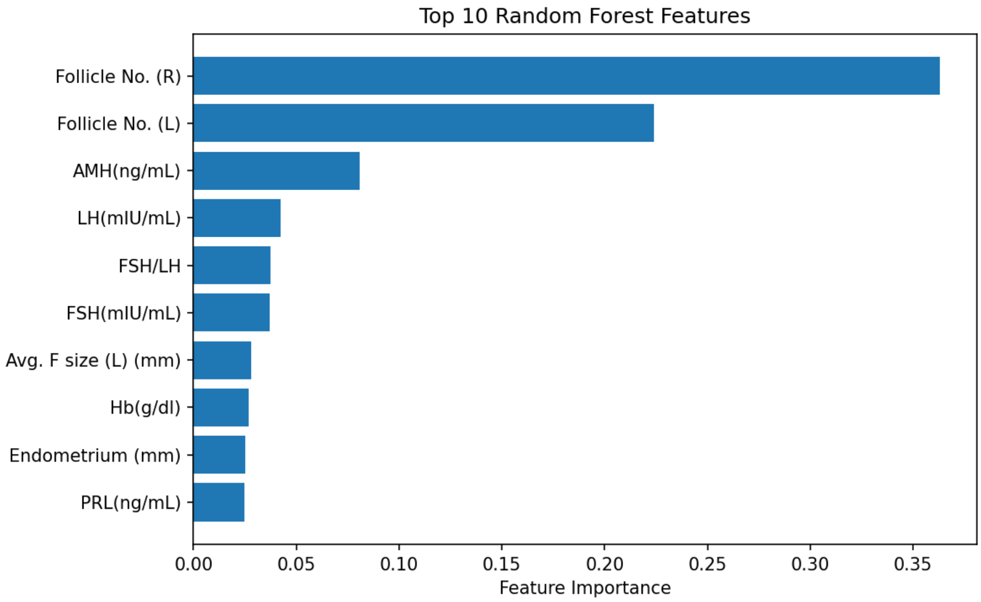

Figure 11: Feature importance scores for the all-minus-doctor-observable feature set obtained using Random Forest.

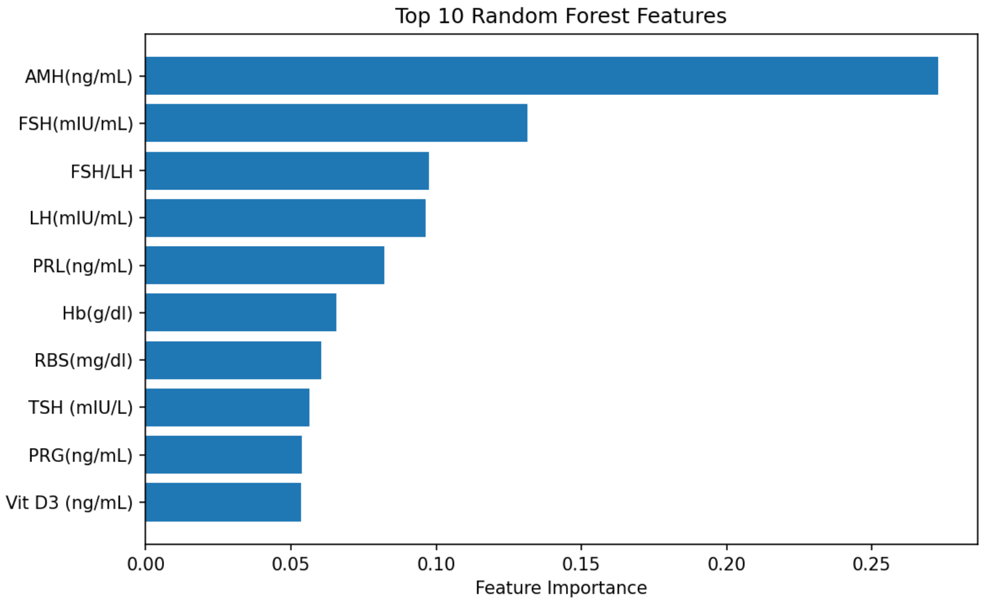

Figure 12: Feature importance scores for the all-minus-gynecological feature set obtained using Random Forest.